

S-Plane Detection — Campaign Validation (v2)

2026-09-20 · Updates [ORAN_SPlane_Master_Test_Catalogue.pdf](#) (items C1-C3) and the Validation-result line of [ORAN_SPlane_Testbed_Configuration_Reference.pdf](#) · Real linuxptp 4.0 G.8275.1 testbed, 6 nodes in network namespaces · Frozen base `decision_rule.py` c362e11..., frozen extension `decision_rule_v2.py` e32eb65b... (both hash-verified unchanged after the run)

What this run is. A randomized campaign of **168 live on-wire runs** — 12 reps × 14 scenarios (11 original + the 3 coverage scenarios C1/C2/C3) — each scored by **both** the frozen rule and the frozen v2 extension on the same runs. Per-rep attacker parameters are randomized (reproducible from the rep number). This is the campaign-grade validation that promotes v2 from a demonstration to a measured result. Clock-safe throughout: netns only, software timestamping, host clock untouched.

Headline (168 runs: 96 attack-expected, 60 benign-expected, 12 abstain-expected)

Metric	Frozen rule	Frozen v2
Attack sensitivity (recall)	0.625 [0.525, 0.715]	1.000 [0.962, 1.000]
Specificity (benign recall)	0.800 [0.682, 0.882]	0.800 [0.682, 0.882]
Abstention correctness (ambiguous → UNKNOWN)	1.000 [0.757, 1.000]	1.000 [0.757, 1.000]

Specificity is **identical** for both rules — v2 is **additive-only**, it adds no false positives and never turns a frozen ATTACK into BENIGN. Both 0.800 figures come entirely from the one documented limitation, `B_bc_replacement` (12/12 flagged, single `expected_bc_identity` in context); the other four benign scenarios (baseline, B2, B3, B7) are **48/48 clean** for both rules.

Per attack class (each n=12)

Attack class	Frozen recall	v2 recall
A1 rogue master · A2 sync spoof · A3 replay · A5 DoS flood · A8 rogue BC	12/12 each (1.000)	12/12 each (1.000)
C1 packet removal / selective interception (O-RAN 11.1.5.3.1)	0/12 (0.000)	12/12 (1.000)
C2 malformed / fuzzed frames (O-RAN 24.2.1.2)	0/12 (0.000)	12/12 (1.000)
C3 whole-second field abuse (IEEE 1588-2019	0/12	12/12

The frozen rule's 0.625 overall sensitivity is not a defect in the five original classes (perfect 60/60) — it is the three coverage gaps scoring 0/36. v2 closes all three, lifting overall sensitivity to 1.000 (96/96) with a 95% lower bound of 0.962.

Document updates carried by this sheet

- **Master Test Catalogue** — items C1/C2/C3, previously "not run", are now campaign-validated (v2 detectors D1/D2/D3, 12/12 each). The frozen rule's blindness to them is confirmed at n=12 each.
- **Testbed Configuration Reference** — its "Validation result" line (81 admissible runs) is superseded: frozen rule 168-run campaign, plus the frozen v2 extension above. The G.8275.1 config itself is unchanged.

■ **Scope & honest limits.** v2 is now campaign-validated (frozen hash e32eb65b, recorded *before* this run; neither rule was tuned to results). Remaining specificity ceiling is the known B_bc_replacement limitation (needs a BC allow-list + re-freeze). Five attack classes stay **hardware-only** and out of software scope: GNSS spoof (A6), GNSS jam (A7), holdover (B1-adjacent), SyncE/EEC (B4), oscillator drift (B6), delay-attack measurement (A4 timing). Evidence: campaign_v3_2026-09-20.tgz (168 runs' deep CSVs + both verdicts + EVALUATION_V3.json + scripts).